A Biologically Plausible Circuit for a Bursting Neuron

Marco Ferroni

***Abstract*—Neuromorphic engineering aims to develop hardware systems that replicate the computational principles of biological neural networks. In this work, we present the design, implementation, and physical layout of a biologically plausible analog circuit that emulates the bursting behavior of cortical neurons. The circuit is based on a minimal model capturing key ionic mechanisms responsible for the generation of burst firing patterns. Our implementation is composed of four distinct feedback mechanisms, whose contributions are joined by the soma block. Each feedback block is independently tunable, allowing for precise control of the bursting dynamics, such as spike frequency, interburst interval, and burst duration. This modular structure ensures flexibility in reproducing a variety of biologically realistic firing patterns while maintaining low power consumption and compact area to achieve stable and controllable bursting activity. The circuit topology is validated through extensive transient simulations, confirming its ability to reproduce typical bursting patterns observed in biological neurons. We also discuss the full custom layout of the neuron circuit in a 180 nm CMOS technology, focusing on the minimization of the area and matching requirements critical for preserving the dynamic behavior. Post-layout simulations, including extracted parasitic effects, demonstrate that the circuit maintains its functional properties, ensuring robustness against process variations and layout-induced mismatches. This work provides a compact and energy-efficient building block for neuromorphic systems, paving the way for the hardware implementation of biologically realistic spiking neural networks.**

I. Introduction

The field of neuromorphic engineering seeks to develop hardware systems that mimic the computational principles and efficiency of biological neural networks. A fundamental feature observed in many types of neurons, particularly in the cortex and thalamus, is bursting — a firing pattern characterized by groups of spikes separated by quiescent periods. Bursting is known to play a critical role in various neural processes, including synaptic plasticity, sensory coding, and signal amplification. Hardware emulation of bursting neurons offers the potential for low-power, high-density, and biologically realistic implementations of spiking neural networks. However, capturing the complex dynamics of bursting in compact and energy-efficient circuits remains a significant challenge. In this work, we present the design, simulation, and physical layout of a compact analog circuit capable of reproducing biologically plausible bursting behavior. The circuit architecture follows a modular approach, featuring four distinct feedback mechanisms whose contributions are integrated by a soma block. Each feedback block is parametrizable, allowing flexible tuning of the bursting dynamics to emulate a variety of firing patterns observed in biological neurons. We validate the proposed design through both pre-layout and post-layout simulations using a 180 nm CMOS process, demonstrating its robustness and low power consumption.

II. Bursting Neuron Model

The proposed model regulates membrane charge through four feedback mechanisms operating at different time scales. These feedback loops alternately add or remove charge, generating the characteristic bursting firing pattern as shown in Fig.1. When a DC input current is applied, the membrane begins to accumulate positive charge, triggering the Fast Positive (FP) feedback mechanism:

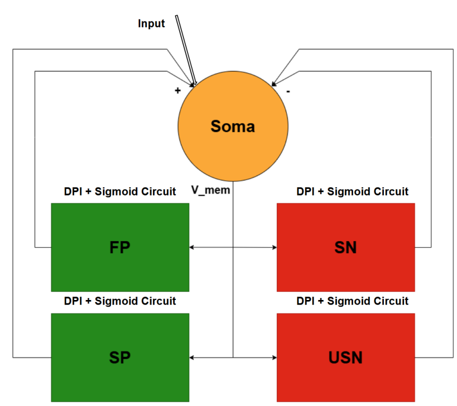

Fig. 1: Biologically compatible model of the neuron

1) Fast Positive Feedback (FP): The Fast Positive Feedback (FP) mechanism enables a rapid influx of positive charge into the membrane, causing a sharp increase in the membrane potential. This fast depolarization mimics the behavior of voltage-gated sodium channels in biological neurons, which are responsible for the initiation and rapid upstroke of action potentials. The activation of FP is essential for triggering individual spikes, as it provides the sudden change in potential needed to cross the firing threshold.

2) Slow Negative Feedback (SN): The Slow Negative Feedback (SN) mechanism acts as a counterbalance to FP by gradually removing charge from the membrane. While it responds more slowly than FP, its influence is stronger and more sustained over time. As the neuron depolarizes due to FP, SN builds up and begins to dominate, reducing the membrane potential and suppressing the spike. This mechanism is analogous to the role of delayed-rectifier potassium channels in real neurons, which contribute to repolarization and the end of an action potential.

3) Slow Positive Feedback (SP): The Slow Positive Feedback (SP) mechanism operates over a longer time scale compared to FP. After a spike has been suppressed by SN, SP remains active and continues to inject charge into the membrane. Once SN becomes sufficiently weak relative to SP, the latter reactivates FP, resulting in a new spike. This repeated interplay between SP and SN leads to the formation of a burst—a train of closely spaced spikes.

4) Ultraslow Negative Feedback (USN): The Ultraslow Negative Feedback (USN) mechanism becomes active during a burst, gradually increasing its influence as the burst progresses. It builds up strength slowly with each spike. Eventually, it becomes strong enough to counteract SP and prevent the reactivation of FP, effectively halting further spiking and terminating the burst. USN helps regulate the overall duration of the burst and brings the neuron back to its resting state. Biologically, it mimics mechanisms such as calcium-activated potassium currents or sodium inactivation, which limit prolonged activity and ensure recovery periods between bursts.

III. Building Blocks

The whole bursting neuron is made of 4 feedback mechanisms and a SOMA circuit that merges their contributions. Each feedback mechanism is formed by the exact same circuit, which is then tuned with the right parameters to achieve the desired behavior; we can divide its structure into two blocks, a sigmoid circuit whose output is connected to a DPI:

1) Sigmoid Circuit: This circuit takes a DC voltage as input and produces a DC current whose response follows a sigmoidal behavior, as shown in Fig.2. This introduces non-linearity into the model and mimics the behavior of ionic channels in real neurons.

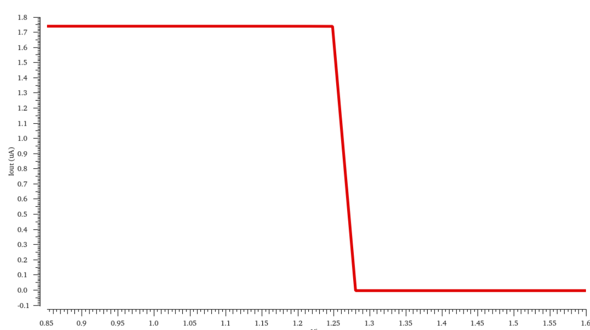

Fig. 2: Output of the sigmoid circuit vs. DC input voltage

The schematic is shown in Fig.3, where mem VP is the input and out UNI is the output. The shape of the sigmoid can be tuned using three parameters: thr VNI, sat VNI, and max VNI. The first controls the input voltage at which the sigmoid begins to transition, the second adjusts the slope of the curve, and the third sets its maximum value.

When a low input voltage is applied, transistor M41 allows a significant current to flow. Transistor M3 can drain part of this current depending on the threshold voltage setting, but if the current exceeds this threshold, it must be discharged through the branch containing M2. In this case, M2 and M43 behave like two forward-biased diodes, creating the sharp transition edge of the sigmoid. Transistor M44 acts as a resistive element, defining the slope at which the common node between M2 and M43 rises with increasing input current.

Finally, the output stage sets the maximum value of the sigmoid through M4, which limits the maximum current allowed to flow in the branch.

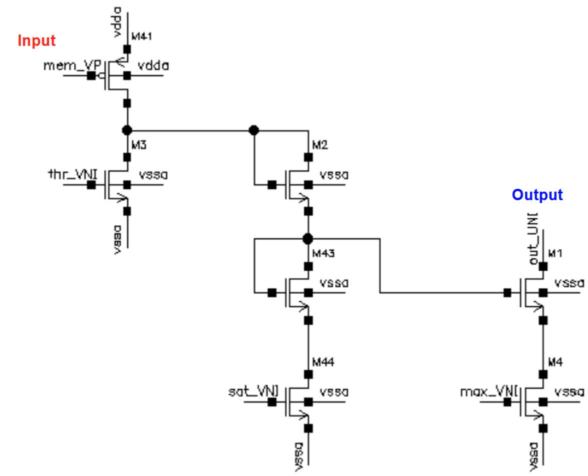

Fig. 3: Schematic of the sigmoid circuit

2) DPI Circuit: This circuit takes a current as input and outputs a voltage that increases with the duration of the input, introducing temporal integration, acting as a form of memory of past signals, as illustrated in Fig.4. Its purpose is to manage the gradual increase in the strength of each feedback mechanism.

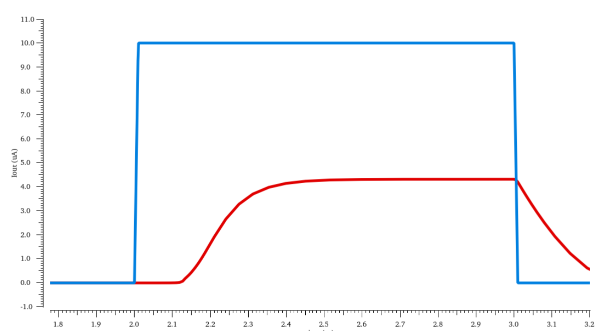

Fig. 4: Output of the DPI circuit vs. square wave input current over time

The schematic is shown in Fig.5, where inp UNI is the input and out UPO is the output. The response can be tuned using two parameters: gain VPI, which sets the maximum output voltage, and tau VPI, which controls the response speed.

When an input current is applied, the branch containing M14 is activated and begins to discharge the cap V0 node. Simultaneously, M25 counteracts this effect, determining the rate at which the node discharges and subsequently recharges once the input current drops to zero. The left branch sets an offset current that is subtracted from the current that needs to be supplied by the right branch, thereby defining the maximum voltage reached by the cap V0 node.

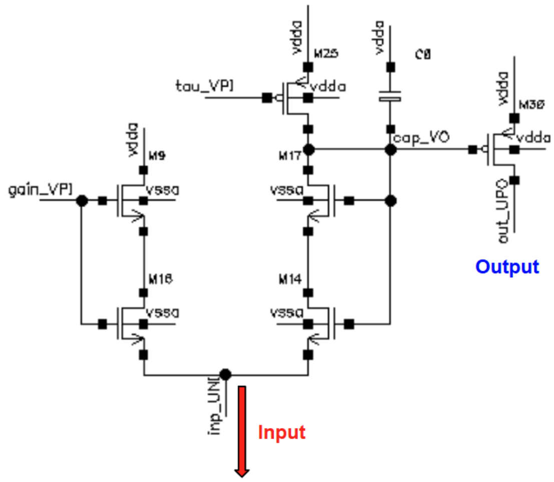

Fig. 5: Schematic of the DPI circuit

3) Soma Circuit: This circuit acts as the core of the neuron; it combines the currents from the four feedback mechanisms and integrates their contributions by modulating them through a time constant (tau) and a gain parameter to adjust the membrane potential. The schematic is shown in Fig.6. It consists of two main blocks: the top one processes the inhibitory inputs (inh UPI), and the bottom one handles the excitatory inputs (exc UPI). Both blocks are essentially DPI circuits that share the same tau transistor and use the same gain setting. The node NN mem VP represents the membrane potential and serves as the input to all feedback mechanisms. A DC current source, located at the bottom left, provides the external input to the entire neuron circuit.

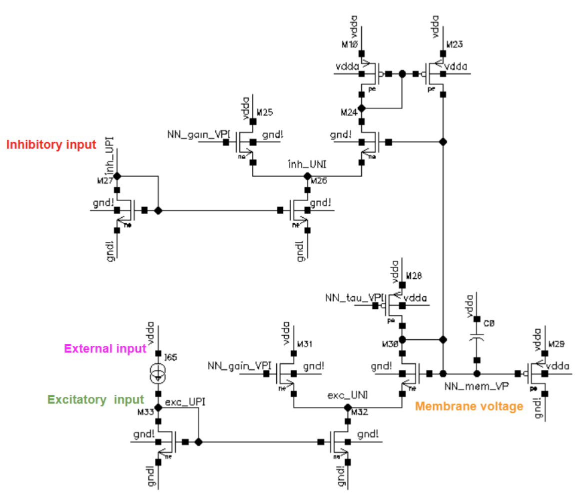

Fig. 6: Schematic of the soma circuit

IV. Parameters tuning and simulations

To verify the functionality of the circuit, we first needed to identify suitable parameters for tuning. To simplify this process, we proceeded step by step.

1) Monotonic Spiking: We initially aimed to achieve monotonic spiking by enabling only the FP and SN feedback mechanisms, as shown in Fig.7.

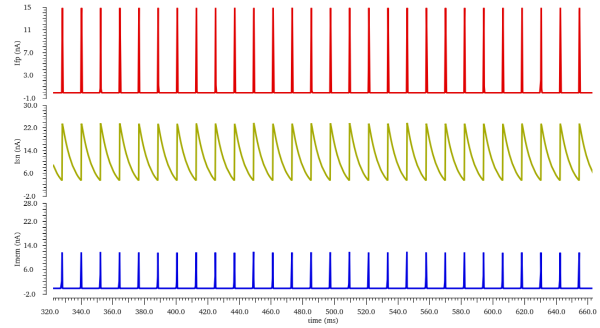

Fig. 7: Monotonic bursting activity of the neuron. The red plot represents FP, the yellow plot represents SN, and the blue plot shows the membrane current.

We also tuned the spiking frequency by modifying the tau of the SN feedback. A larger tau causes SN to decay more quickly after it has ”killed” a spike, allowing a new one to occur sooner, as shown in Fig.8.

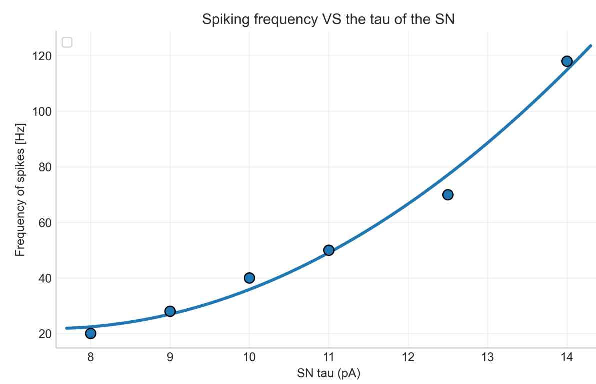

Fig. 8: Spiking frequency vs. the tau of the SN

2) Bursting: Following this, we introduced the SP and USN mechanisms to generate bursting activity, as illustrated in Fig.9.

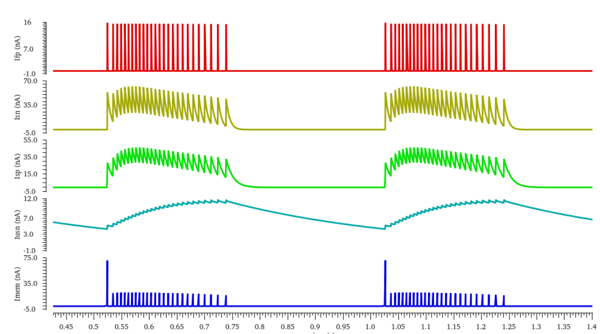

Fig. 9: Bursting activity of the neuron. The green plot corresponds to SP, while the light blue plot corresponds to USN.

Just as we did for monotonic spiking, we also tuned the bursting frequency, this time by adjusting the tau of the USN. The principle is the same: a larger tau causes USN to lose strength more quickly, allowing a new burst to begin sooner, as shown in Fig.10.

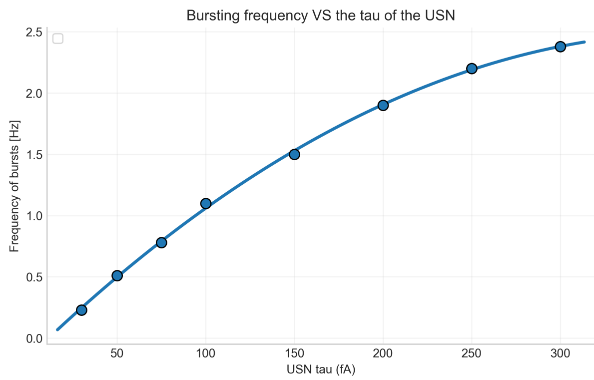

Fig. 10: Bursting frequency vs. the tau of the USN

V. Pre-layout Simulations

To compare the behavior of the circuit before and after layout, we evaluated both its robustness and power consumption.

1) Power Consumption: To evaluate power efficiency, we measured the energy required to generate a single spike and analyzed its variation with the tau of the SN feedback. We found that increasing tau leads to reduced energy consumption, as shown in Fig.11. This is because the spike is terminated more quickly, reducing its duration and thus the energy it consumes.

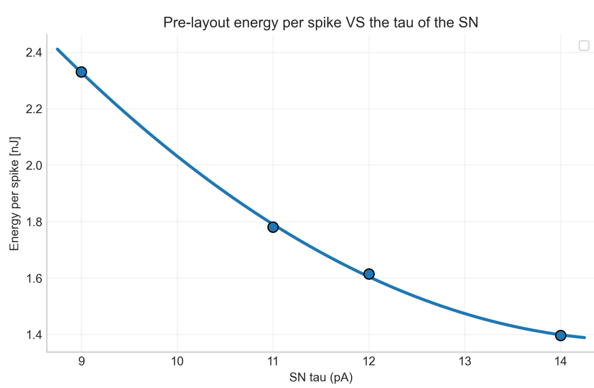

Fig. 11: Pre-layout energy per spike vs. the tau of the SN

2) Monte Carlo Simulation: Given the circuit's sensitivity to parameter selection, we evaluated its robustness to both process variations and mismatch. We used the average output current as an indicator of proper functionality; when the circuit failed to spike, the average current increased (in absolute value), serving as a diagnostic signal. After initial simulations, we identified transistors contributing most to mismatch and increased their sizes, while reducing the non-critical ones to save area. With the optimized sizing, we achieved an 86% success rate, as shown in Fig.12.

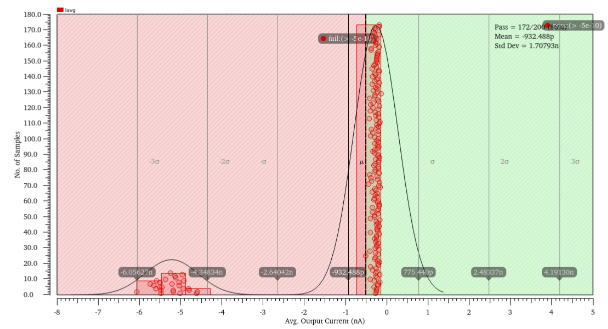

Fig. 12: Pre-layout Monte Carlo simulation for mismatch and process variations

VI. Physical Layout

Once the minimum functional transistor sizes were identified, we proceeded with layout implementation.

1) Feedback Block Layout: We began with a single feedback block. PMOS and NMOS transistors were separated using dummy devices to ensure uniform boundary conditions, with space reserved for guard rings around each cluster once they were part of the final layout. To save area, we used a capacitor whose plates were placed on unused metal layers (METTP and MET5), allowing vertical stacking. The asymmetric placement of transistors and capacitor on one side was intentional: the final layout places the four feedback blocks in a way to create a central region reserved for the soma. The layout of a single feedback block is shown in Fig.13.

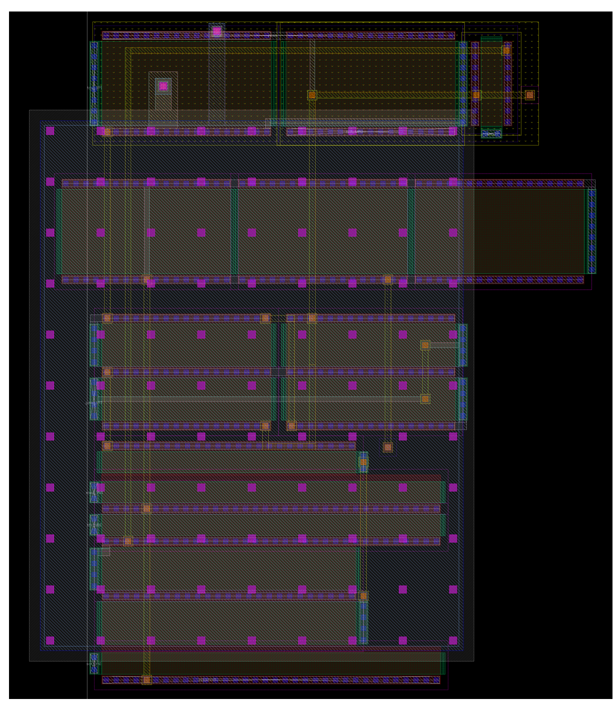

Fig. 13: Layout of a single feedback block

2) Final Layout: The layout of the bursting neuron instantiates all four feedback blocks, mirrored to form a central cavity where the soma is placed. The resulting structure features PMOS rows at the top and bottom, and a single NMOS block at the center. Guard rings were added to prevent latch-up: two

P-guard rings for the PMOS stripes and one large N-guard ring for the NMOS. Additional dummy devices were placed around the entire circuit to ensure uniform boundary conditions in any instantiation of the block. A power grid was then created to distribute power efficiently across the circuit with low impedance. The total layout area is $75.66 \times 67.02\,\mu\text{m}^2$. Fig.14 shows the final layout.

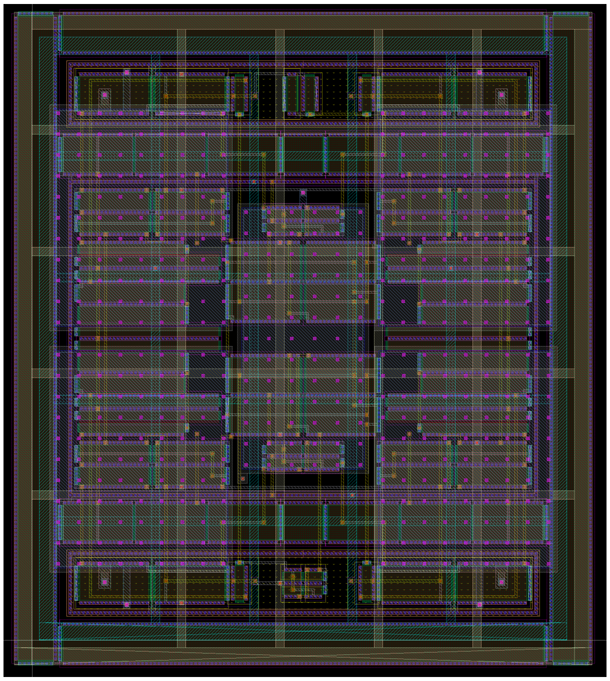

Fig. 14: Final layout of the neuron

VII. Post-layout Simulations

The next step was to assess the quality of the layout; first we generated the extracted view and verified that the circuit continued to function as intended. Then we repeated both the Monte Carlo analysis and power evaluation.

1) Power Consumption: As with the pre-layout analysis, we measured the energy required to generate a spike. As expected, post-layout energy consumption was higher due to parasitic effects as shown in Fig.15.

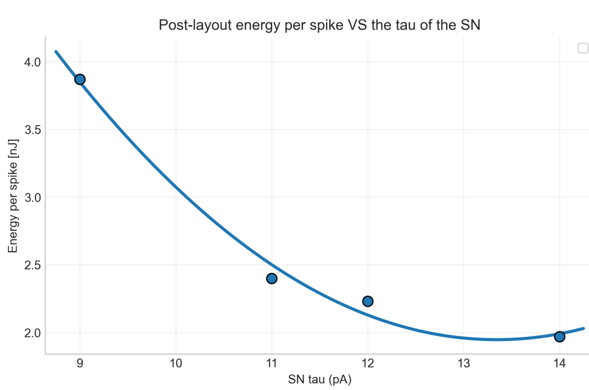

Fig. 15: Post-layout energy per spike vs. the tau of the SN

2) Monte Carlo Simulation: To evaluate post-layout robustness, we performed a new Monte Carlo simulation. This run was computationally intensive and took approximately 10 hours to complete. The final yield was 77%, as shown in Fig.16.

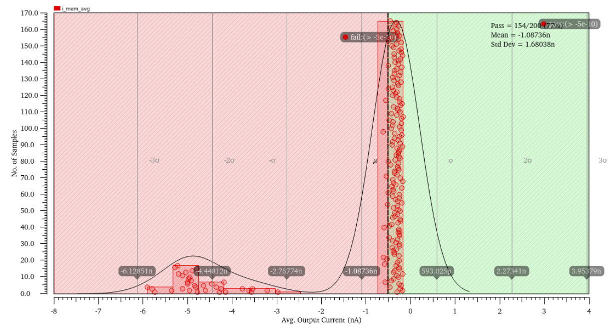

Fig. 16: Post-layout Monte Carlo simulation for mismatch and process variations

VIII. Conclusions

In this work, we designed and implemented a neuromorphic circuit capable of producing biologically-inspired spiking and bursting activity through the interaction of four feedback mechanisms. We explored the tunability of spiking and bursting frequencies by adjusting time constants, and analyzed power consumption and robustness both before and after layout. Transistor sizing was optimized to improve yield while minimizing area, without significantly compromising power efficiency. Post-layout simulations confirmed the circuit's functionality, showing only moderate performance degradation due to parasitics. The final design demonstrates a robust and compact neuron implementation suitable for integration in larger neuromorphic architectures.